

# Analysis

Dr. Duff Baker-Jarvis

May 14, 2026

**Definition 1** (Compactness). A set  $S \subset U$  (in a topological space) is called **compact** if for every open cover  $\mathcal{T}$  of  $S$  there is a finite subcover of  $S$ .

**Theorem 2.** The image of a compact set under a continuous function is compact.

Proof idea: If  $f : D \rightarrow \mathbb{R}$  is a continuous function and  $D$  is compact, then let  $\mathcal{T}$  be any open cover of  $f(D)$ . Then taking the preimage of each  $T \in \mathcal{T}$  gives an open cover of  $D$ . This has a finite subcover,  $\mathcal{T}_0$ . Then mapping these back to elements of  $\mathcal{T}$  gives a finite subcover of  $f(D)$ .

**Theorem 3.** Let  $S$  be a compact set on  $\mathbb{R}$  and  $f : S \rightarrow \mathbb{R}$  be continuous. Then  $f$  achieves both its maximum and minimum on  $S$ .

Proof: By 2 theorem,  $f(S)$  is compact. By Heine-Borel, it is closed and bounded and hence  $\sup f(S) = \max f(S)$  and likewise for the minimum.  $\square$

**Theorem 4** (Rolle's Theorem). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function continuous on  $[a, b]$  and differentiable on  $(a, b)$  such that  $f(a) = f(b)$ . Then there exists  $c \in (a, b)$  s.t.  $f'(c) = 0$ .

**Theorem 5** (Mean Value Theorem). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function that is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists a  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof: Consider the function  $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$ . Then  $g(a) = f(a)$  and  $g(b) = f(a)$ . Thus by Rolle's Theorem there exists a  $c \in (a, b)$  s.t.  $g'(c) = 0$  and the rest follows.  $\square$

**Theorem 6** (Taylor's Theorem). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function that has its first  $n$  derivatives continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Fix a point to expand around,  $x_0$ . Let  $x \in [a, b]$  with  $x \neq x_0$ . Then for any  $n \in \mathbb{N}$  there exists and  $c \in (x, x_0)$  such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

Idea of Proof:

- Construct a function  $F(t)$  such that

1.  $F$  is continuous on  $(a, b)$  and differentiable on  $[a, b]$ .
2.  $F(x) = f(x)$  and  $F(x_0) = f(x_0)$
3.  $F'(t) = 0$  forces the correct coefficient of  $(x - x_0)^{n+1}$

- Use the mean value theorem on  $F(t)$ .

Noteworthy property of Modified Taylor Series: If  $P(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x - t)^k$  then

$$P'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n$$

Proof:( following Lay 28.2) Let  $x \in [a, b]$  and fix  $n \in \mathbb{N}$ . Let

$$F(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x - t)^k + M(x - t)^{n+1}$$

where  $M$  is the unique solution to

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + M(x - x_0)^{n+1}.$$

Note that by the noteworthy property mentioned above we have

$$F'(t) = \frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n + 1)M(x - t)^n.$$

Note also that  $F(x) = F(x_0) = f(x)$ , so invoking the mean value theorem (or Rolle's Theorem) we obtain that there exists a  $c$  between  $x$  and  $x_0$  strictly such that

$$F'(t) = 0 = \frac{f^{(n+1)}(t)}{n!}(x - t)^n + (n + 1)M(x - t)^n.$$

Solving for  $M$  gives  $M = \frac{f^{(n+1)}(x_0)}{(n+1)!}$  as required.  $\square$

**Example 7.** Let  $f(x) = x^2 + 2x + 1$  on  $[0, 3]$  with  $x_0 = 0$ . Let  $n = 1$ . Then there should exist a  $c$  such that

$$f(2) = 1 + 2(x - 0) + \frac{f^{(n+1)}(c)}{(n + 1)!}(x - 0)^{n+1}$$

$$1 + 2x + \frac{2}{2}x^2$$

so we recover our polynomial.

We now note that we can rephrase this theorem in terms of a remainder.

**Theorem 8** (Taylor's Remainder Theorem). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be  $n + 1$  times differentiable and  $x_0 \in \mathbb{R}$ . Then if we approximate

$$f(x) \approx p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

then there exists a  $c$  between  $x$  and  $x_0$  such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}.$$

Note though, that  $c$  depends on  $x$ .

**Remark 9.** Note that if  $f$  is a polynomial of degree  $k$ , then  $p_n = f$  for all  $n \geq k$  and all  $x_0 \in \mathbb{R}$ .

**Lemma 10.** Let  $f$  be differentiable in a neighborhood of  $c$ . Then

$$f''(c) = \lim_{h \rightarrow 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2}$$

$$f''(c) = \lim_{h \rightarrow 0} \frac{(f(c+h) - f(c)) + (f(c-h) - f(c))}{h^2}$$

Proof: Starting with the limit and using L'Hospital's rule we get:

$$\begin{aligned} f''(c) &= \lim_{h \rightarrow 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} = \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c-h) - 2f'(c)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c-h) + 2f'(c-h) - 2f'(c)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{1}{2} \frac{f'(c+h) - f'(c-h)}{h} + 2 \lim_{h \rightarrow 0} \frac{f'(c-h) - f'(c)}{2h} \end{aligned}$$